5

Probabilities

Chapter Learning Outcomes

CLO

5

- Distinguish between experiment, sample space and event.
- List all possible outcomes in the sample space of a random experiment.
- Understand the probability of an event.
- Determine the probability of some events.
- Determine the probability of the union and intersection of two or more events.
- Determine the probability of the complement of an event.
- Distinguish between mutually exclusive (disjoint) and inclusive (joint) events..
- Distinguish between independent and dependent events.
- Use Venn Diagrams to depict single events, complementary events, unions and intersections of a collection of events.
- Calculate conditional probabilities.
- Apply probability rules in solving probability problems.

Probability

- **Experiment**

  An action, or trial, through which specific results are obtained.

- **Sample Space**

  The set of all possible outcomes of a probability experiment.

- **Event**

  Consists of one or more outcomes and is a subset of the sample space.

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EXAMPLE

Experiment: Roll a dice

Sample space:

Event: {Die is even}=

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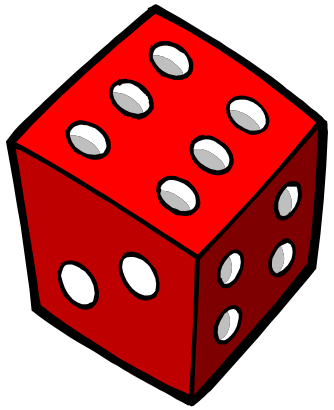
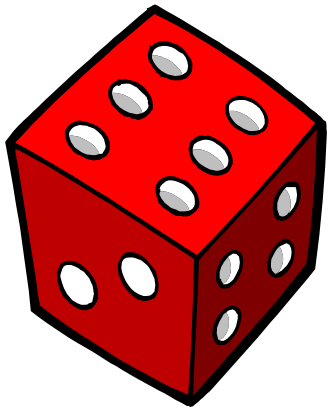
EXAMPLE

Experiment: Tossing a coin and then rolling a dice.

Describe the sample space.



Probability



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Probability of an Event

$$P(E) = \frac{\text{Number of outcomes in event } E}{\text{Number of outcomes in sample space } S}$$

$$P(E) = \frac{|E|}{|S|}$$

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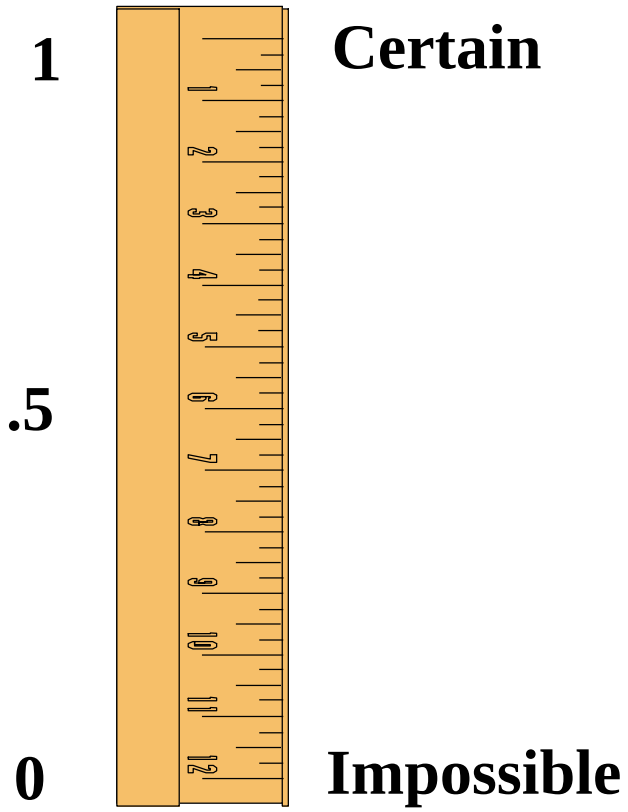
Range of Probabilities Rule

$$P(E) = \frac{|E|}{|S|}$$

$$0 \le P(E) \le 1$$



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Probability

EXAMPLE

You roll dice. Find the probability of each of the following events:

1. *Event A: rolling a 3*
2. *Event B: rolling a 7*
3. *Event C: rolling a number less than 5*

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Probability

EXAMPLE

What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7.

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EXAMPLE

A box contains 4 blue balls and 5 red balls. What is the probability that a ball chosen from the box is blue?

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EXAMPLE

A special die is made in the shape of an octahedron. The die has 8 equal faces marked with the numbers 1 to 8. If the die is thrown once, what is the probability that the face that lands uppermost has a prime number?

Solution:

Sample space (S) :

Event (E) :

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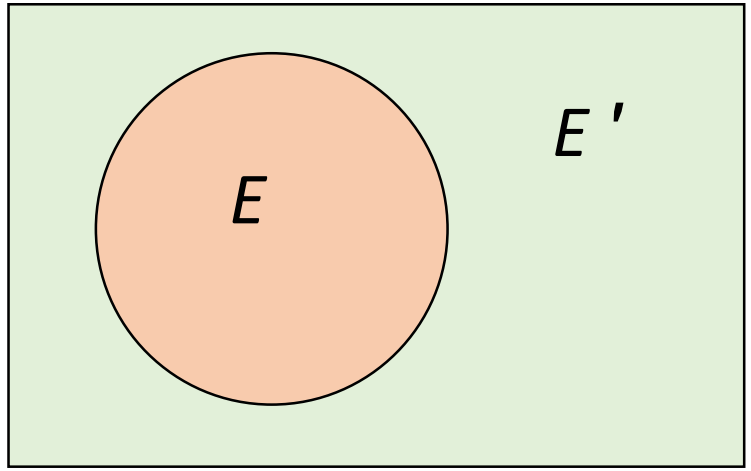
Probability - Rules

1- Complement of Event (E)

$$P(E') = 1 - P(E)$$



Remember!

$$P(E) + P(E') = 1$$

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Probability - Rules

EXAMPLE

Assume P(A)=0.6, P(B)=0.3, find:

1. P(A').
2. P(B').

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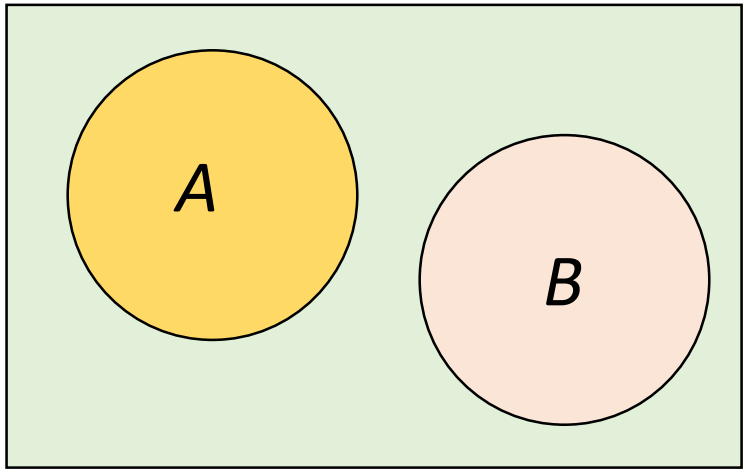
2- Disjoint Events (Mutually Exclusive)



$$P(A \cup B) = P(A) + P(B)$$

$$P(A \cap B) = 0$$

Remember!

Recall that the union symbol $\cup$ means "or"

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Probability - Rules

EXAMPLE

A group of students is donating blood during a blood drive. A student has a $\frac{9}{20}$ probability of having type O blood and a $\frac{2}{5}$ probability of having type A blood.

What is the probability that a student has type O or type A blood?

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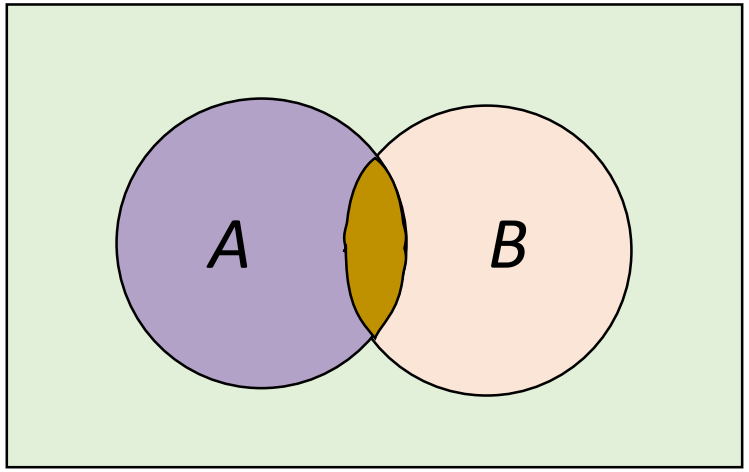
Probability - Rules

3- Joint Events (Inclusive)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Remember!

Recall that the intersection symbol $\cap$ means "and"

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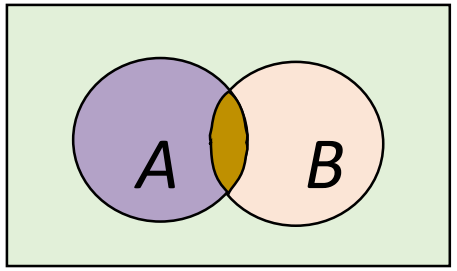
Probability - Rules

3- Joint Events (Inclusive)



4- Independent Events

$$P(A \cap B) = P(A) \times P(B)$$

Definition

In Independent events, the outcome of one event <u>does not affect</u> the outcome of the other

5- Dependent Events

$$P(A \cap B) = P(A) \times P(B|A)$$

Definition

In dependent events, the outcome of one event affects the outcome of the other

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Probability - Rules

EXAMPLE

A coin is tossed and a dice is rolled. Find the probability of getting a head on a coin and 6 on a dice.

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Probability - Rules

EXAMPLE

Two cards are selected, without replacing the first card, from a standard deck. Find the probability of selecting a king and then selecting a queen.

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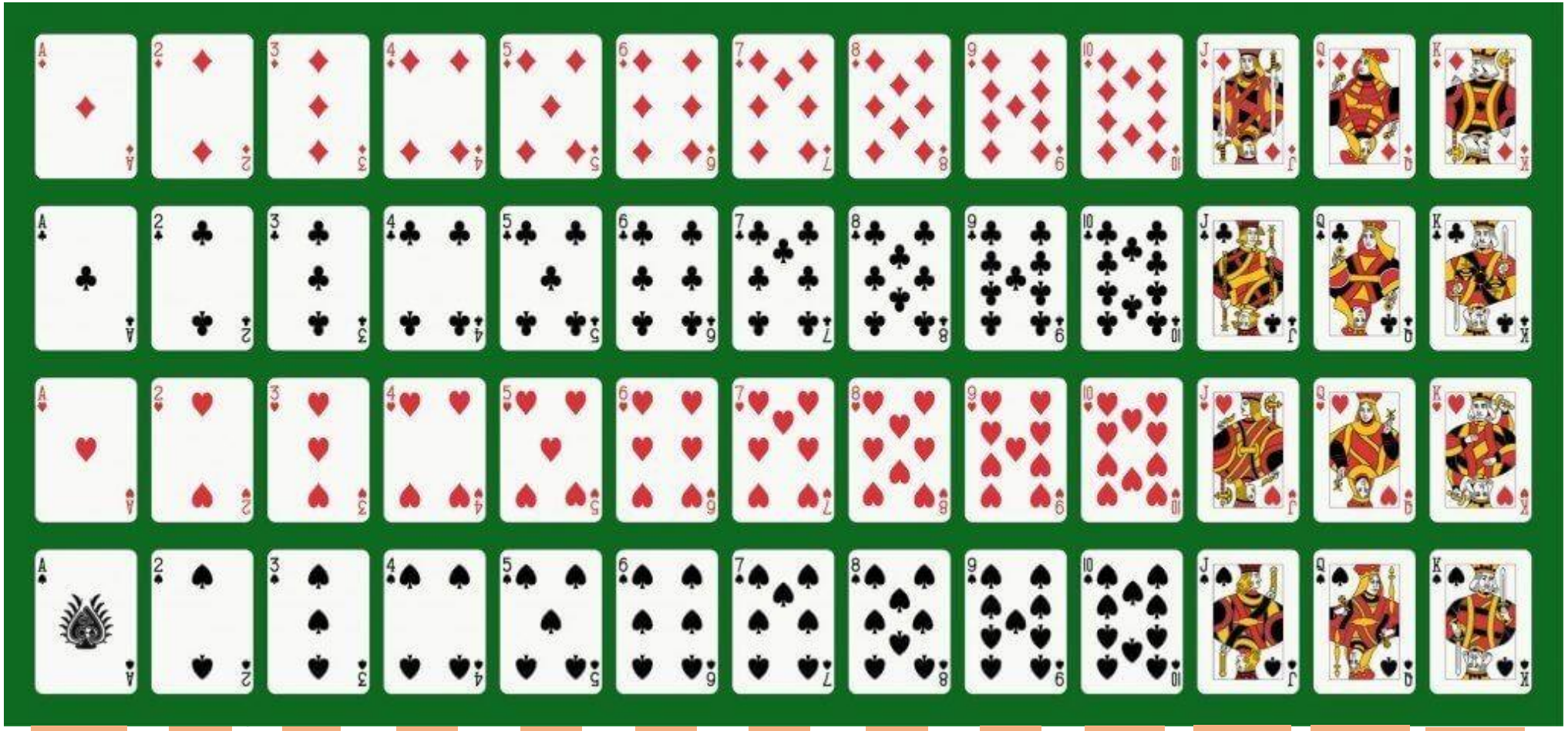
Standard Deck

Diamond
ديناري

Club
سنك

Heart
كبة

Spade
بستوني



Ace | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Jack | Queen | King

6- Conditional Probability

Definition

- **The probability of an event occurring, given that another event has already occurred.**
- **Denoted P(B|A) (read "probability of B, given A")**

$$P(A|B) = P(A \cap B) \div P(B)$$

$$P(B|A) = P(A \cap B) \div P(A)$$

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Probability – Rules Summary

1- Complement of Event (E)	$P(A') = 1 - P(A)$
2- Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
3- Disjoint Events (Mutually Exclusive)	$P(A \cap B) = 0$
4- Joint and Independent Events	$P(A \cap B) = P(A) \times P(B)$
5- Joint and Dependent Events	$P(A \cap B) = P(A) \times P(B\|A)$
6- Conditional Events	$P(A\|B) = P(A \cap B) \div P(B)$

17 Examples

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Example 1

EXAMPLE

Assume $P(A) = 0.3$, $P(B) = 0.6$, $P(A \cup B)=0.8$

1. Find $P(A \cap B)$.
2. Are A and B disjoint events.
3. Find $P(A|B)$.
4. Find $P(B|A)$.

1. Find $P(A \cap B)$.

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Example 1

EXAMPLE

Assume $P(A) = 0.3$, $P(B) = 0.6$, $P(A \cup B)=0.8$

1. Find $P(A \cap B)$.
2. Are A and B disjoint events.
3. Find $P(A|B)$.
4. Find $P(B|A)$.

2. Are A and B disjoint events.

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Example 1

EXAMPLE

Assume $P(A) = 0.3$, $P(B) = 0.6$, $P(A \cup B)=0.8$

1. Find $P(A \cap B)$.
2. Are A and B disjoint events.
3. Find $P(A|B)$.
4. Find $P(B|A)$.

3. Find $P(A|B)$.

4. Find $P(B|A)$.

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Example 2

EXAMPLE

One card is drawn from the deck.

1. Find the probability of randomly selecting number 2.
2. Find the probability of randomly selecting a face.

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Example 3

EXAMPLE

Of 1560 students surveyed, 840 were females. 630 read a daily newspaper. The rest of the students were males. Only 215 of the newspaper readers were males. **What is the probability that a student was a female or read a daily newspaper?**

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Example 4

EXAMPLE

Of 570 people, 365 were male. 368 had black hair. Of those with black hair, 108 were female. **What is the probability that a person was male or had black hair?**

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Example 5

EXAMPLE

The table shows the favorite pizza topping for a sample of students. One of these students is selected at random. **Find the probability the student is male, given that they prefer Onion**.

	Tomato	Onion	Mushroom	Total
Male	8	5	2	15
Female	2	4	3	9
Total	10	9	5	24

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Example 6

EXAMPLE

The table shows the favorite pizza topping for a sample of students. One of these students is selected at random. **Find the probability the student is female or prefers mushroom.**

	Tomato	Onion	Mushroom	Total
Male	8	5	2	15
Female	2	4	3	9
Total	10	9	5	24

Example 7

EXAMPLE

You survey a sample of 1000 employees at a company and record the age of each.

1. Find the probability of randomly choosing an employee who is between 25 and 34 years old.
2. Find the probability of randomly choosing an employee who is <u>not</u> between 25 and 34 years old.

Employee ages	Frequency, f
15 to 24	54
25 to 34	366
35 to 44	233
45 to 54	180
55 to 64	125
65 and over	42
	$\Sigma f = 1000$

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Example 7

Employee ages	Frequency, f
15 to 24	54
25 to 34	366
35 to 44	233
45 to 54	180
55 to 64	125
65 and over	42
	$\Sigma f = 1000$

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Example 8

EXAMPLE

Each student cast one vote for senior class president. Of the students, 25% voted for Ali, 20% for Laith, and 55% for Salem. A student from the senior class is selected at random.

What is the probability that a student voted for:

1. Laith?
2. Ali?
3. Salem?
4. Laith and Salem?
5. Laith or Salem?

Name	Prob., P
Ali	0.25
Laith	0.20
Salem	0.55
	$\Sigma P = 1$

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Example 9

EXAMPLE

One dice is rolling. Find the probability of rolling a 4 or an even number

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Example 10

EXAMPLE

One dice is rolling. Find the probability of rolling an odd number or a number greater than 2

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Example 11

EXAMPLE

A one card is drawn from a deck of 52. Find the probability of drawing a king or a heart

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Example 12

EXAMPLE

The probability that a particular knee surgery is successful is 0.85. Find the probability that three knee surgeries are successful.

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Example 13

EXAMPLE

The probability that a particular knee surgery is successful is 0.85. Find the probability that none of the three knee surgeries is successful.

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Example 14

EXAMPLE

True or False:

The following events are mutually exclusive.

Event A: Being born in Irbid
Event B: Studying at BAU.

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Example 15

EXAMPLE

- **Assume the probability of getting a flu is 0.2.**
- **Assume the probability of having a high fever given the flu: 0.9.**

Find the probability of getting a flu with fever?

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Example 16

EXAMPLE

- **Assume the probability of getting a flu is 0.2.**
- **Assume the probability of getting a fever is 0.3**
- **Assume the probability of having a high fever given the flu: 0.9.**

Find the probability of having a flu given the fever?

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Example 17

EXAMPLE

- **Assume the probability of getting a flu is 0.2.**
- **Assume the probability of getting a fever is 0.3**
- **Assume the probability of having a high fever given the flu: 0.9.**

Are flu and fever independent ?

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Check - Chapter Learning Outcomes

CLO

5

- Distinguish between experiment, sample space and event.
- List all possible outcomes in the sample space of a random experiment.
- Understand the probability of an event.
- Determine the probability of some events.
- Determine the probability of the union and intersection of two or more events.
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- Distinguish between mutually exclusive (disjoint) and inclusive (joint) events..
- Distinguish between independent and dependent events.
- Use Venn Diagrams to depict single events, complementary events, unions and intersections of a collection of events.
- Calculate conditional probabilities.
- Apply probability rules in solving probability problems.

End of Chapter

5